

ONE-SIDED KAZHDAN TRANSPORT AND LEAVITT SELF-COMPRESSION

SAUERS

A countable group is MF if it embeds in the unitary group of a norm matrix corona. We prove a one-sided Kazhdan transport theorem for operator-norm asymptotic representations and apply it to the explicit group

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$$

over the binary Leavitt algebra. The group is nontrivial, finitely generated, simple, and has property (T). An explicit Whitehead matrix compresses a Kazhdan subgroup into itself, and the resulting distinguished involution converges to the identity in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation. Every finite-dimensional linear representation of H over every field is trivial.

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CONSUMPTION**

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1. INTRODUCTION

A countable group is *MF* if it embeds in the unitary group of a norm matrix corona

$$\mathcal{Q}_d = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \rightarrow 0\}$$

is the c_0 -direct sum. Equivalently, an MF group admits finite-dimensional unitary models whose multiplicative defects tend to zero in operator norm and which asymptotically separate its nonidentity elements. Throughout, *MF group* means this operator-norm notion of Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor]. A stronger regular-norm approximation property is also called MF in parts of the recent literature [GKEMP]; it is not the convention used here.

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The existence of a countable non-MF group remains a central operator-norm approximation problem [Sh]. The results below isolate an exact one-sided transport phenomenon relevant to that problem. We do not infer an operator-norm obstruction from the normalized Hilbert–Schmidt conclusion.

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(1) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem A. *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{12}(R)$. Then H is nontrivial, simple, finitely generated, and has property (T). There is a nonidentity involution $d \in H$ such that every operator-norm asymptotic representation (V_n) of H satisfies*

$$\|V_n(d) - 1\|_2 \longrightarrow 0.$$

Moreover, every finite-dimensional linear representation of H over every field is trivial.

Suppose that $L \leq G$ has property (T), that $u \in G$ satisfies $uLu^{-1} \leq L$, and that c centralizes L . For every operator-norm asymptotic representation (V_n) of G ,

$$(2) \quad \left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Inside H , an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the upper-left subgroup $L \cong \text{EL}_3(R)$. The element $c = e_{34}(1)$ centralizes L , and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The element d is nontrivial and H is simple, so d normally generates H . Since the coefficient ring has characteristic two, d is an involution.

For an exact finite-dimensional representation, conjugation by a compressor defines an injective endomorphism of the commutant of L . This endomorphism is surjective because the commutant is finite-dimensional. Consequently, every finite-dimensional linear representation of H over every field is trivial.

The argument is specific to operator-norm approximation. It does not prove that H is nonsofic or nonhyperlinear: normalized Hilbert–Schmidt approximations do not provide the operator-norm control used to construct the conjugation representation in Theorem 3.3.

Relation to prior work. The group-MF framework used here is developed by Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor]; recent permanence results are due to Shulman [Sh]. The operator-norm case remained open after non-approximability had been established for the finite Schatten norms. Bachner–Dogon–Lubotzky emphasize the interaction between

operator-norm defects and Hilbert–Schmidt rigidity as a possible route to the MF problem [BDL]. The transport theorem below concerns that interaction, but its normalized Hilbert–Schmidt conclusion is not promoted to an operator-norm conclusion here.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser’s normal-subgroup theorem implies simplicity of the elementary group [Pre], while Ershov–Jaikin–Zapirain give property (T) for $\text{EL}_n(R)$ whenever $n \geq 3$ and R is finitely generated [EJZ]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. ONE-SIDED COMPRESSION IN FINITE DIMENSION

Theorem 2.1 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k -vector space, and $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\text{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. It commutes with every $\rho(\ell)$, $\ell \in L$, which proves the last assertion. $\square$

3. KAZHDAN TRANSPORT IN NORMALIZED HILBERT–SCHMIDT NORM

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator-norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \quad (g, h \in G).$$

Lemma 3.1. *For every sequence (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v . Then $x_n^*x_n \rightarrow 1$ in norm. For all sufficiently large n , the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. In a stably finite unital C^* -algebra, equivalent projections $p \leq q$ satisfy $p = q$. $\square$

Lemma 3.2 (one-sided order for the Kazhdan projection). *Let L have property (T), let B be a unital C^* -algebra, and let $\pi: L \rightarrow \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C_{\max}^*(L) \rightarrow B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\text{Fix } \pi(L)$. If ξ is L -fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B . $\square$

Theorem 3.3 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator-norm asymptotic representation of G , and let (x_n) be uniformly bounded in operator norm. If*

$$\|V_n(\ell)x_nV_n(\ell)^* - x_n\|_2 \longrightarrow 0 \quad (\ell \in L),$$

then, for every $\ell \in L$,

$$\|V_n(\ell)V_n(u)x_nV_n(u)^*V_n(\ell)^* - V_n(u)x_nV_n(u)^*\|_2 \longrightarrow 0,$$

$$\|V_n(\ell)V_n(u)^*x_nV_n(u)V_n(\ell)^* - V_n(u)^*x_nV_n(u)\|_2 \longrightarrow 0.$$

Proof. Let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the Hilbert-space ultraproduct $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the operator-norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0 . The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

The preceding estimate shows directly that $g \mapsto \tilde{\sigma}(g)$ is an exact homomorphism $G \rightarrow \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$ -homomorphism $C_{\max}^*(L) \rightarrow \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$ [BHV, Chapter 3]. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$. Hence both U and U^* preserve $\text{Fix } \sigma(L)$, so $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. The two limits in the statement therefore vanish along ω . Since this holds for every free ultrafilter, both sequences converge ordinarily. $\square$

Corollary 3.4. *Let $L \leq G$ have property (T), let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. For every operator-norm asymptotic representation (V_n) of G and every $\ell \in L$,*

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0.$$

Proof. Let (V_n) be an operator-norm asymptotic representation. If $c \in C_G(L)$, the sequence $(V_n(c))$ asymptotically commutes with $V_n(L)$ in operator norm and hence in normalized Hilbert–Schmidt norm. Theorem 3.3 shows that the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$. Therefore

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \quad (\ell \in L).$$

$\square$

4. THE BINARY LEAVITT SELF-COMPRESSION

Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0 t_0, \quad q = s_1 t_1.$$

The relations (1) give

$$(3) \quad p + q = 1, \quad t_1 q s_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices and identify $\text{GL}_3(R)$ with the subgroup of $\text{GL}_{12}(R)$ consisting of the matrices $\text{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$(4) \quad \Psi(A) = qI_3 + s_0 A t_0.$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q s_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0 \Psi(A) s_0 = A$

proves injectivity. The term qI_3 is needed for unitality because the map $A \mapsto s_0At_0$ sends I_3 to pI_3 .

Set

$$U = \begin{pmatrix} s_0I_3 & s_1t_0I_3 \\ 0 & t_1I_3 \end{pmatrix}, \quad V = \begin{pmatrix} t_0I_3 & 0 \\ s_0t_1I_3 & s_1I_3 \end{pmatrix}.$$

A direct calculation using (1) gives $UV = VU = I_6$. Put

$$(5) \quad \tau = \text{diag}(U, V) \in \text{GL}_{12}(R).$$

Writing $I = I_6$, the Whitehead factorization is

$$(6) \quad \text{diag}(U, U^{-1}) = \begin{pmatrix} I & U \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -U^{-1} & I \end{pmatrix} \begin{pmatrix} I & U \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

Each block-unipotent factor is a product of elementary 12×12 matrices: its off-diagonal 6×6 block may be inserted one entry at a time, and the corresponding matrix units have pairwise zero products. Since $V = U^{-1}$, (6) proves $\tau \in \text{EL}_{12}(R)$.

Let $H = \text{EL}_{12}(R)$ and let $L = \text{EL}_3(R) \leq H$ be the upper-left corner. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ]. Block multiplication yields

$$(7) \quad \tau \text{diag}(A, I_9) \tau^{-1} = \text{diag}(\Psi(A), I_9).$$

For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0at_0)$. Therefore

$$(8) \quad \tau L \tau^{-1} \leq L.$$

5. THE LEAVITT GROUP

Proposition 5.1. *The group $H = \text{EL}_{12}(R)$ is simple.*

Proof. The ring $R = L_{\mathbb{F}_2}(1, 2)$ is purely infinite simple [AA] and therefore an exchange ring [Ara]. Let $N \trianglelefteq H$. Preusser's normal-subgroup theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I),$$

where $C_{12}(R, I)$ is the full congruence subgroup of level I . Since R is simple, either $I = R$ or $I = 0$. In the first case $\text{EL}_{12}(R, I) = H$, so $N = H$. In the second case

$$N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R)).$$

The center of R is $\mathbb{F}_2$ by [AC, Corollary 4.3]. For completeness, suppose that $z \in Z(\text{GL}_{12}(R))$. Commuting z with the elementary matrices $e_{ij}(1)$ shows that $z = \lambda I_{12}$ for some $\lambda \in R$. Commuting with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times = \mathbb{F}_2^\times = \{1\}$. Thus $Z(\text{GL}_{12}(R))$ is trivial, and $N = 1$ when $I = 0$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 5.2. *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad d^2 = 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. Every elementary generator of the upper-left 3×3 corner has both indices in $\{0, 1, 2\}$. The elementary commutator relations show that it commutes with $c = e_{34}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (5) gives

$$(9) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with $\ell = e_{12}(1)$, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$. This is nontrivial because $q \neq 0$. Since R has characteristic two and $E_{02}^2 = 0$, one also has $e_{02}(q)^2 = 1$. Finally, d is nontrivial and H is simple by Proposition 5.1, so its normal closure is all of H . $\square$

Proof of Theorem A. The group H is nontrivial and simple by Proposition 5.1. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ]; in particular H is finitely generated.

The containment (8), the centrality of c relative to L , and Corollary 3.4 show that every operator-norm asymptotic representation (V_n) of H satisfies

$$\|V_n(d) - 1\|_2 \longrightarrow 0.$$

Now let k be a field, let V be finite-dimensional over k , and let $\rho: H \rightarrow \mathrm{GL}(V)$ be a homomorphism. The containment (8), the centrality of c relative to L , and Theorem 2.1 give $\rho(d) = 1$. Proposition 5.2 says that d normally generates H , so ρ is trivial. $\square$

In particular, every homomorphism from H to a finite group, a residually finite group, or a compact Hausdorff group is trivial. For finite and residually finite targets this follows by the regular representation of a finite quotient; for compact targets it follows from Peter–Weyl point separation [PW].

AI AND COMPUTATIONAL RESOURCE USAGE

AI systems were used extensively for proof exploration, literature navigation, formalization, and editorial revision. Selected supporting lemmas and related variants have been formalized in Lean; this does not constitute an end-to-end formal verification of the present manuscript. The author checked and revised the mathematical arguments and citations.

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